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The Effect of Music on Dental Anxiety

A literature Review submitted by:

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Outline:

I. Introduction

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2. Theoretical background: the effect of dental anxiety, factors that cause dental anxiety and importance of music
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2. Studies with children: (Dixit & Jasani, 2020; Tshiswaka & Pinheoro, 2020).
 - a) Purpose & Questions/Hypothesis.
 - b) Sample
 - c) Instruments and procedures
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 - e) Practical implications

3. Studies with adults: (Gupta & Ahmed, 2020; Packyanathan et al., 2020).

- a) Purpose & Questions/hypothesis.
- b) Sample
- c) Instruments and procedures
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IV. Discussion:

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question: The four studies showed positive effect of music on dental anxiety.

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- Repeating each experiment more than once
- Experimenting on a different age group (adolescents)

References

I. Introduction:

A major problem faced by patients undergoing dental treatment is dental anxiety, a highly prevalent condition affecting about one in six patients, including 6.3% to 43% of youth (Dixit & Jasani, 2020) and roughly 12% of adults (Gupta & Ahmed, 2020). It is commonly triggered by factors such as fear of pain, clinical sounds, and past negative experiences. As a result, this intense worry often leads patients to avoid appointments and delay necessary care, which in turn contributes to poorer oral health and creates more stressful treatment conditions for dental professionals (Packyanathan et al., 2019; Tshiswaka & Pinheiro, 2020). While dental professionals have traditionally relied on sedatives or general anesthesia to manage severe patient anxiety, these pharmacological interventions carry significant drawbacks—such as drowsiness, blood pressure fluctuations, and higher costs—highlighting the demand for effective, nonpharmacological alternatives. (Gupta & Ahmed, 2020). One promising non-pharmacological approach is music therapy. Music can reduce anxiety by moderating heart rate, lowering stress hormones, and triggering the neurological release of dopamine and serotonin, which improves mood and decreases pain perception (Tshiswaka & Pinheiro, 2020). Its anxiety-reducing effects are well-documented in general medical procedures, such as Cesarean sections and surgical biopsies (Dixit & Jasani, 2020; Packyanathan et al., 2019). However, despite this evidence, research specifically examining the impact of music on dental anxiety remains limited (Gupta & Ahmed, 2020).

To address this gap in the literature, the purpose of this review is to determine the overall effect of music on dental anxiety. This objective is guided by the research question: what is the impact of music on dental anxiety?

In this review, music (the independent variable) is defined in one study as instrumental earth sounds (Gupta & Ahmed, 2020). In another study, music is categorized as children's songs in Portuguese (Tshiswaka & Pinheiro, 2020). In a third study, it is defined as the classical music of Veena (Packyanathan et al., 2019). In the last study, music is defined as Indian classical instrumental music of Raag Sohni played by Pandit Shiv Kumar

Sharma on santoor (Dixit & Jasani, 2020). Dental anxiety (the dependent variable) is also defined in various ways. In one research study, dental anxiety is defined by the heart rate measurements that were taken before, after, and during the dental care treatment, and also by a score of state anxiety during the treatment (Tshiswaka & Pinheiro, 2020). In another study, dental anxiety is represented by a score of heart rate variation before, during, and after treatment, and by a score of state anxiety from 0 -10, such that 0 means no anxiety and 10 means the highest level of anxiety (Gupta & Ahmed, 2020). In the third study, dental anxiety is defined as scores of heart rate variation and changes in blood pressure, together with a score ranging from 5 to 25 showing how anxious the patient is as the score increases (Packyanathan et al., 2019). In the last study, dental anxiety is defined operationally by a score of state anxiety by depicting happy and unhappy faces, the number of observed disruptive behaviors exhibited by the patient, and a score of dental anxiety from 15 – 75 (the highest score) with 38 the benchmark for the presence of anxiety (Dixit & Jasani, 2020).

II. Methodology

Scopus database was used to find the research studies included in this review. The keywords that were used were *dental anxiety* and *music*. Some delimiters were used including PDF full text, research studies written in English, and the years were limited from 2011 to 2021 which gave 253 results. To further refine the results, the search was limited to the 'Abstract' field for the terms 'Music' and 'Dental anxiety.' Additionally, the search was restricted to the 'Medicine' subject area, yielding a final sample of 16 results.

Out of the 16 research studies, 5 were excluded because they were secondary. Six of the studies were primary; however, they were excluded because they were irrelevant to the research question. One study did not include patients, and examined the effect of music practice on dental students' stress levels. Another study evaluated the effect of the behavior management techniques and the use of Nitrous oxide to decrease the anxiety levels of adult pediatric patients. The last study evaluated the impact of a novel communication aid *Message to Dentist* (MTD) on dental anxiety. One study was excluded because it was written in Spanish, and two of the primary irrelevant studies were repeated.

The remaining three studies were primary relevant studies so they were included in this literature review (Dixit & Jasani, 2020; Packyanathan et al., 2019; Tshiswaka & Pinheoro, 2020). One study, conducted by Gupta and Ahmed (2020), was a supplementary study obtained from Google Scholar.

III. Results:

All four articles included in this literature review can be grouped into two groups according to the age of the sample. The first group contains the studies which have children. This group includes two studies (Dixit & Jasani, 2020; Tshiswaka & Pinheoro, 2020). The second group contains studies which have adults. This group includes (Gupta & Ahmed, 2020; Packyanathan et al., 2019).

The first study belonging to the first group and conducted by Dixit and Jasani (2020) *aimed* at comparing the impact of Bach Flower Therapy and Music Therapy on the minimization of dental anxiety in children. The researchers' *research question* can be inferred as: what is the impact of Bach flower therapy and music therapy on the decrease of children's dental anxiety? As for the *research sample*, the study included a convenient sample of 120 children aged from 4-6 from India; 73 participants were males and the rest were females. Regarding the *instruments*, to measure anxiety, the North Carolina Behavior Rating Scale (NCBRS) was used. For completing this scale, an independent assessor observed the treatment video and noted the displayed disruptive behavior of the patient out of 4 identified behaviors. In addition, the Facial Image Scale was used; it included 5 faces (ranging from very unhappy to very happy) from which the child chose the one that best described his/her current feeling. The pulse Oximeter was also used to measure the pulse rate and oxygen saturation of all children. Finally, the Manual sphygmomanometer and stethoscope was used to measure systolic and diastolic blood pressure. Regarding the *procedure*, it was a pre-post design. The chosen children were divided equally into 3 groups; music therapy group (MT), Flower therapy group (BFT), and control group. Before treatment, the Facial Image Scale was completed and the physiological measurements were evaluated by a trained assistant. 15 minutes before the treatment, the BFT group took four drops of "rescue remedy" diluted in 40 ml of water, however, the MT group and the control group took 40 ml of plain water. During the treatment, the BFT and control group were told to wear headphones without playing music; on the contrary, the MT group were given headphones which had Indian classical instrumental music playing (Raag Sohni played by Pandit Shiv Kumar Sharma on santoor). During the treatment, the pulse rate, oxygen saturation, and systolic

and diastolic blood pressure were measured. After the treatment, the NCBRS and the Facial Image Scale were completed. The *results* of this study showed that the behavior of children in the BFT group was better than those in the control group ($P=0.014$). In the intraoperative time period, the pulse rate of the participants in both BFT and MT groups decreased ($p<0.001$) compared to the control group which significantly increased ($P < 0.001$). In addition, diastolic blood pressure in the decreased in MT and BFT groups and increased in the control group. At the postoperative time period, the frequency of the children in different FIS categories showed no statistically significant difference across the three groups ($P=0.243$). However, blood pressure in children in the MT group was lower than that of children in both the BFT and control group and the postoperative time period. There were no *practical implications* included in this study.

The second study belonging to the first group and conducted by Tshiswasaka and Pinheiro (2020) *aimed* at determining the impact of music on children's anxiety during dental treatment. The researchers' *hypothesis* was music does not impact children's anxiety level during dental treatment. The *sample* consisted of 40 children between the ages of 5-11 from the Children's clinic of the school of Dentistry of PUC-Campinas, Brazil. As for the participants' gender, 23 were males and the rest were females. The selection criterion was convenient. Regarding the *instruments*, oxygen saturation and heart rate were measured with a finger oximeter (G-TECH, Model Oled graph, Beijing, China). A pain scale was also included; the children were asked to note the image that best described their feeling during and after the treatment. To measure state anxiety, the Corah Dental Anxiety Scale was used which required children to rate their feelings at the moment of contacting the assistants and the dentist by answering 4 questions with 5 possible answers. Regarding the *procedure*, the children were divided randomly into 2 groups: One music group and the other non-music group. Before the welcome and after the treatment, the children completed the pain scale and the corah dental anxiety scale, then the welcome started by explaining the procedure of the research and the treatment to the children. Before the reception, during and after treatment, the oxygen saturation and heart rate were measured. For the music group, the children went to the treatment rooms, and handsets with mp3 songs and headphones were distributed to the children after being seated before the beginning of the treatment. The same repertoire heard by the children was connected to the sound in the room. For the control group, after welcoming them, they were led to the treatment rooms and had the treatment without listening to music. The type of procedure was pre-post design. The *results* showed that the heart rate (pulse) of the children who listened to music during dental treatment decreased ($p=0.05$). The heart rate (pulse) of the children who did not listen to music during dental treatment did not change during the treatment ($p=0.53$). Therefore, the hypothesis was rejected. However, there was not a noticeable change in oxygen saturation, Corah anxiety scale and pain for the children who listened to music during dental treatment. As for *practical implications*, the researchers did not mention them.

After having a look at the studies with children, the studies including adults will be tackled.

The first study belonging to the second group and conducted by Gupta and Ahmed (2020) *aimed* at examining how music impacts anxiety and pain levels of dental patients during minor oral surgery (MOS) procedures. The *research question* can be inferred as: What is the impact of music on patient anxiety levels during MOS procedures? The researchers included a convenient *sampling* of 50 adult participants aged 20-80 years old, who were attending the MOS clinic at Birmingham Dental Hospital from 25 June 2016 till 25 July 2016. 68% of the patients were females, 32% were males. As for the *instruments*, a 0–10-point scale was used to measure patients' anxiety levels, and a pulse oximeter measured the heart rate variation. In addition, a short questionnaire evaluated the patients' overall experience: whether music had an impact on their anxiety and pain levels and how and whether they would like music if they were to have the same treatment again. The researchers used a pre-post *procedure*. Anxiety levels were measured before and after the treatment, and the heart rate changes were noted before, during, and after the treatment. Patients were handed earphones connected to an iPad including instrumental music with earth sounds. The surgical procedure was then conducted while the patients listened to music. After the treatment, the patients filled the short questionnaire. For *results*, 92% of the patients stated that music decreased their anxiety, and 90% of the patients wanted to listen to music during their next visit. Patients also reported that music made communication easier and reduced pain and discomfort. In addition, heart rate was reduced greatly during the procedure. As for *practical implications*, they were not mentioned.

In the last study belonging to the second group that was conducted by Packyanathan et al. (2019), it *aimed* to examine how music therapy affects dental anxiety in patients during their extraction procedure. The *research question* of this study can be inferred as: What effect does music therapy have on patients undergoing a dental extraction procedure? The *sample* included 50 patients aged between 17 and 64. Participants needed to be healthy and fit for the extraction, and participants who had any systemic disease or were pregnant or deaf were excluded from the study. The location was the department of oral and Maxillofacial surgery at Saveetha

Dental collage, Chennai, India. As for the selection criterion, it was not stated. Regarding the *instruments*, anxiety levels were measured using the Modified Dental Anxiety Scale, which is a brief scale of 5 items. Heart rate changes and blood pressure were also measured using a pulse oximeter. Concerning the *procedure*, after the consent was taken from the patients, they were divided into 2 groups (test group and control group). All participants were then asked to wear headphones. For the test group, classical music was played in their headphones for 5 minutes before and during the extraction procedure. The control group did not hear any music, and their headphones were only on noise cancellation mode. The anxiety levels were measured before and after the treatment following a pre-post procedure design. Regarding *results*, the music group witnessed a significant decrease in systolic pressure, diastolic pressure, and heart rate, whereas for the control group, these parameters increased (P value < 0.05). The Modified Dental Anxiety Scale showed a significant reduction in anxiety levels for the music group in comparison to the control group whose anxiety levels increased (P value < 0.05). For *practical implications*, the researchers recommended the use of music in the waiting area through headphones to reduce anxiety prior to any dental treatment.

IV. Discussion:

- *Overview of results related to the examined research question*
- *Differences across the 4 studies*
- *Interpretation of results*
- *Recommendations for future research*

Declaration of Generative AI Use

I acknowledge the use of NotebookLM (Google, 2026) to assist in the thematic synthesis and organization of the literature review section of this paper. This tool was used to identify common themes and structure the relationships between the sources. All content was subsequently reviewed, edited, and verified for accuracy by the author.

During the preparation of this work the author used Scopus AI (Elsevier, 2026) in order to discover relevant peer-reviewed literature for the background section. After using this tool/service, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

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